Pre-work · The Digestive System

The Alimentary Canal and The Accessory Digestive Organs. Read both Part A chapters first. The canal is one continuous tube with one wall plan and a short list of local exceptions, so learn the plan before you learn the regions, or you will memorize the same four layers six times.

Module 4 · 2 of 5 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the two digestive chapters into mini visuals. Eight chunks. Two chains carry the two routes, the food and the bile, and those two are the ones you will be asked to trace.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 **SPLIT**

The canal against the accessory organs

One split. On one branch the organs food actually passes through, on the other the six organs that stay outside the tube and deliver into it. Write beside each accessory organ how it reaches the lumen, by direct contact or by a duct.

2 **LADDER**

The four layers of the wall

Four rungs, mucosa at the top down to serosa. Beside the mucosa rung write its own three sublayers, and beside the muscularis rung write its two directions of smooth muscle. Then mark the two exceptions: where the serosa becomes an adventitia, and where the muscularis gains a third layer.

3 **CHAIN**

The pathway of food, mouth to anus

Seven boxes with arrows, mouth to anus. Under each box write in three words what happens to the food there, and mark every sphincter it has to pass on the way out.

4 **HUB**

The stomach

One J-shaped stomach at the hub. Spoke to the cardia, fundus, body, and pyloric part, the two curvatures, the sphincter at each end, the rugae, and the gastric pits with their glands. Draw the three muscle layers as three arrows pointing three different ways.

5 **LADDER**

Three levels of surface amplification

Three rungs, the largest fold at the top and the smallest at the bottom: circular folds, villi, microvilli. Beside each write which layers of the wall it is built from, and hang the lacteal and the capillary bed off the villus rung.

6 **HUB**

The large intestine

One large intestine at the hub, drawn as a frame around the small intestine. Spoke to the cecum and appendix, the four parts of the colon in order, the rectum, and the anal canal with its two sphincters. Then mark the three features that name the colon on sight: teniae coli, haustra, and omental appendices.

7 **HUB**

The liver, from lobe to lobule

The liver at the hub. Spoke out to the four lobes, the falciform ligament, and the ligamentum teres in its free edge, then spoke inward to one lobule: hepatocytes in plates, sinusoids running between them, bile canaliculi, and a portal triad at the corner. Write beside the triad which two members bring blood in and which one carries bile out.

8 **CHAIN**

The path of bile, canaliculus to duodenum

Six boxes with arrows, bile canaliculi through to the duodenum. Draw the gallbladder as a side branch off the cystic duct rather than as a box in the line, because bile only detours there. Mark the ampulla and the sphincter that guards it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

The abdominal organs and the folds of peritoneum

BRAIN DUMP FIRST

Two minutes. The tube in place, the membranes that hold it, and which organs sit behind them.

Draw the abdomen in sagittal section with the diaphragm across the top. Lay in the stomach, the small intestine, and the colon, then draw the parietal peritoneum along the wall and the visceral peritoneum over the organs, with the peritoneal cavity between them. Add the greater omentum hanging as an apron from the greater curvature, the lesser omentum between the stomach and the liver, the falciform ligament running to the anterior wall, the mesentery holding the jejunum and ileum, and the mesocolon. Then shade the retroperitoneal organs behind the parietal layer and label each one.

In your notebook, head the page **M4-7 The abdominal organs and the folds of peritoneum** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say for any abdominal organ whether it lies inside the peritoneal cavity or behind it, and name the fold that anchors it.

DRAWING 2

The stomach, cut open

BRAIN DUMP FIRST

One minute. Every named region and border, and the wall in cross section.

Draw the stomach J-shaped with the anterior wall cut away. Mark the cardia at the esophagus, the fundus bulging above and to the left, the body, and the pyloric part with its antrum, canal, and pylorus. Label the lesser curvature along the concave border and the greater curvature along the convex one, and put the lower esophageal sphincter and the pyloric sphincter at the two ends. Draw the rugae running through the empty stomach, then take a small square of the wall and enlarge it to show the four layers with three muscle layers instead of two, and a gastric pit leading down into a gastric gland.

In your notebook, head the page **M4-8 The stomach, cut open** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your own drawing why the stomach carries a third muscle layer that the rest of the tube does not.

DRAWING 3

The liver, gallbladder, pancreas, and the duct tree

BRAIN DUMP FIRST

Two minutes. Three organs and every duct that joins them to the duodenum.

Draw the liver with its right and left lobes and the falciform ligament between them, the gallbladder tucked under the right lobe, the duodenum curving below, and the pancreas behind, with its head in that curve and then its body and tail. Now draw the ducts: the right and left hepatic ducts joining as the common hepatic duct, the cystic duct running out to the gallbladder, those two forming the common bile duct, the pancreatic duct joining it at the hepatopancreatic ampulla, and the accessory duct entering the duodenum on its own. Mark the major duodenal papilla and the sphincter of Oddi.

In your notebook, head the page **M4-9 The liver, gallbladder, pancreas, and the duct tree** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can put one gallstone anywhere on your duct tree and say exactly what backs up behind it.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient has yellow skin and eyes, pale stools, and an inflamed pancreas. Imaging finds a single stone. Name the one place it is lodged and explain why one stone produces all three findings.

Answer. In the hepatopancreatic ampulla.

Reasoning. Work backward from what is blocked rather than forward from the stone. Yellow skin and pale stools say that bile is not reaching the duodenum, so the obstruction sits somewhere on the bile path. An inflamed pancreas says that pancreatic juice is not leaving either, so the obstruction is also on the pancreatic path. A stone in the cystic duct would trap bile inside the gallbladder and leave the pancreas alone, and a stone in the common hepatic duct would sit above the point where the pancreatic duct joins, so neither one fits both findings. Only a single place on the map belongs to both routes: the ampulla where the common bile duct and the pancreatic duct meet, just before the major duodenal papilla. One stone, one shared passage, two organs backed up behind it.

Set A · Name it

1. The nerve plexus lying between the two muscle layers, which controls the movement of the gut, is the ________________________.
2. The fatty apron of peritoneum hanging from the greater curvature of the stomach is the ________________________.
3. The lymphatic capillary inside each intestinal villus is the ________________________.
4. The three thickened bands of longitudinal muscle running the length of the colon are the ________________________.
5. Name the three members of a portal triad and say which way each one carries its fluid.

Set B · Predict it

1. The lower esophageal sphincter weakens. Predict the symptom and name the condition.
2. The gallbladder is removed. Predict what happens to bile, working from the path it takes without that detour.
3. A peptic ulcer erodes through the last layer of the stomach wall. Predict what follows and name every layer it crossed.
4. Disease flattens the circular folds and villi of the small intestine. Predict the effect and explain it from surface area.

Set C · Tell it apart

1. The esophagus is the one region of the canal without a serosa. Say what it has instead and why the difference matters if it is perforated.
2. Distinguish the exocrine tissue of the pancreas from its endocrine tissue by structure, product, and route out.
3. Explain why an inflamed appendix hurts in the lower right abdomen while a weak esophageal sphincter is felt behind the breastbone.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M4-

The canal against the accessory organs

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M4-2

The four layers of the wall

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M4-3

The pathway of food, mouth to anus

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

M4-4

The stomach

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M4-5

Three levels of surface amplification

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M4-6

The large intestine

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M4-7

The liver, from lobe to lobule

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M4-8

The path of bile, canaliculus to duodenum

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.